

The Basics of Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation, or RAG, is a technique developed to optimize the outputs of Large Language Models (LLMs). Standard LLMs are prone to hallucinations and often lack access to specific, private, or real-time external data. RAG solves this by fetching relevant document snippets and injecting them directly into the AI prompt layout.

The Architecture and Processing Workflow:

1. **Text Chunking:** The system reads a PDF document and splits long passages into smaller semantic units, typically around 500 characters with a 50-character overlap margin, to preserve context.
2. **Embedding Generation:** Each text chunk is converted into high-dimensional numerical vectors using the `gemini-embedding-001` model to represent semantic meaning.
3. **Database Indexing:** These vectors are safely serialized and indexed locally inside an SQLite relational database.
4. **Semantic Retrieval:** When a user submits a query, the system vectorizes the question and uses Cosine Similarity to mathematically identify the top 3 most relevant segments.

Language Model Capabilities:

For the final text generation, the system utilizes the `gemini-2.5-flash` model. Gemini relies heavily on advanced Transformer architectures for Natural Language Processing (NLP). If a user asks a question that cannot be answered using the provided context, the AI is instructed to reply with a polite apology instead of speculating.